

BOIZENBURG, Germany

WMO: 102490

Lat: 53.3911N

Lon: 10.6878E

Elev: 46

StdP: 100.77

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.4	-9.7	-16.2	0.9	-11.3	-13.6	1.2	-9.4	13.0	5.7	11.5	6.2	2.7	30	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.5	18.5	26.6	17.8	24.6	16.9	19.5	26.1	18.5	24.8	17.6	23.3	4.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.4	22.1	16.1	11.5	21.2	15.0	10.7	20.1	55.5	26.3	52.5	24.9	49.5	23.3	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.1	8.1	-13.1	31.8	5.6	2.4	-17.1	33.5	-20.4	34.9	-23.5	36.2	-27.5	38.0		
(5)			-13.7	22.0	5.4	0.8	-17.5	22.5	-20.7	23.0	-23.7	23.5	-27.6	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	0.3	0.5	4.1	7.6	12.2	15.2	17.4	17.3	13.8	9.6	4.5	2.2		
	DBStd	7.17	5.74	4.78	3.71	3.29	3.64	3.37	3.02	2.89	2.55	3.53	3.36	3.87		
	HDD10.0	1316	299	267	185	86	18	1	0	0	2	49	166	242		
	HDD18.3	3553	558	499	442	323	192	104	54	56	138	272	415	500		
	CDD10.0	870	0	1	2	14	87	158	229	226	116	37	1	0		
	CDD18.3	66	0	0	0	0	2	12	26	23	2	0	0	0		
	CDH23.3	650	0	0	0	1	36	129	271	196	17	1	0	0		
	CDH26.7	142	0	0	0	0	3	27	68	43	1	0	0	0		
(14) Wind	WSAvg	4.0	4.8	4.4	4.4	3.8	3.6	3.3	3.4	3.2	3.7	4.0	4.3	4.6		
(15) Precipitation	PrecAvg	712	63	50	49	41	54	69	79	75	55	54	59	64		
	PrecMax	949	121	124	114	89	109	150	176	182	164	142	116	150		
	PrecMin	521	2	8	6	2	10	14	10	26	13	26	4	29		
	PrecStd	114	32	25	29	20	25	33	40	39	33	26	28	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.3	16.8	21.2	26.7	30.0	30.9	30.1	25.7	21.4	13.4	11.9		
		MCWB	8.8	10.6	10.7	13.4	16.7	19.3	18.8	19.2	18.1	16.9	10.4	9.5		
	2%	DB	9.3	10.8	13.1	18.3	24.1	27.2	28.6	28.0	22.5	18.2	11.8	10.2		
		MCWB	7.3	8.0	8.9	11.7	15.3	18.2	18.7	18.3	16.6	14.0	9.7	8.5		
	5%	DB	8.0	8.2	10.8	15.9	22.1	24.6	26.7	25.7	20.1	16.1	10.6	8.9		
		MCWB	6.1	6.0	7.5	10.1	14.5	16.5	18.1	17.6	15.0	12.6	8.7	7.3		
	10%	DB	6.8	6.3	9.3	13.6	19.6	22.2	24.3	23.4	18.3	14.5	9.2	7.7		
		MCWB	5.1	4.3	6.7	9.0	13.6	15.5	17.1	16.7	13.9	11.6	7.7	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	10.7	11.0	14.3	17.5	20.9	21.2	20.5	18.9	17.1	11.3	10.2		
		MCDB	10.6	14.0	16.0	19.9	24.9	27.0	28.0	27.6	23.8	21.1	12.5	11.5		
	2%	WB	7.6	7.9	9.3	12.1	16.2	19.0	19.6	19.0	17.0	14.6	9.9	8.8		
		MCDB	9.2	10.3	12.5	16.9	22.6	25.1	26.4	25.5	21.2	17.4	11.3	10.0		
	5%	WB	6.4	6.1	8.1	10.7	15.0	17.4	18.5	18.2	15.9	13.1	8.9	7.5		
		MCDB	7.9	8.1	10.4	14.8	20.8	22.5	25.4	24.1	19.3	15.6	10.3	8.8		
	10%	WB	5.2	4.4	6.8	9.5	13.7	16.1	17.4	17.2	14.6	11.9	7.8	6.4		
		MCDB	6.7	6.1	9.0	13.3	19.1	21.1	23.6	22.6	17.6	14.2	9.3	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	6.5	8.7	9.7	9.3	9.8	9.5	8.0	6.8	4.7	4.0		
		MCDBR	5.0	7.1	9.7	12.8	13.8	13.9	14.2	13.7	10.9	9.0	6.3	5.2		
	5% WB	MCWBR	4.5	5.7	6.7	7.7	7.4	6.8	6.8	6.6	6.4	6.2	5.3	4.8		
		MCDWR	4.8	7.0	8.8	11.2	12.5	12.1	13.0	11.4	9.7	8.4	6.0	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.337	0.369	0.410	0.406	0.403	0.422	0.415	0.385	0.356	0.333	0.308		
		taud	2.346	2.344	2.280	2.211	2.265	2.294	2.257	2.290	2.361	2.420	2.393	2.340		
	Ebn at Noon		652	758	807	817	843	848	822	805	784	728	627	586		
		Edh at Noon	59	80	104	126	126	124	126	116	96	74	56	49		
(44) All-Sky Solar Radiation	RadAvg		0.58	1.27	2.43	4.00	5.02	5.30	4.99	4.15	3.05	1.68	0.72	0.44		
	RadStd		0.07	0.17	0.24	0.54	0.63	0.41	0.72	0.33	0.34	0.20	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-144	N/A		

Nomenclature: See separate page